C.W
20/07

Problem Set 1.3

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Limits Problem:-

⑧ $\lim_{x\to\infty} \frac{3x-2}{2x+1} = \frac{\lim_{x\to\infty} x\left(3-\frac{2}{x}\right)}{x\left(2+\frac{1}{x}\right)} = \frac{3}{2}$

⑨ $\lim_{x\to\infty} \frac{x^2}{\sqrt{x^4+1}} = \lim_{x\to\infty} \frac{x^2}{\sqrt{x^4\left(1+\frac{1}{x^4}\right)}} = \lim_{x\to\infty} \frac{x^2}{x^2\sqrt{1+0}} = 1$

⑪ $\lim_{x\to\infty} \frac{\sqrt{x+3x^2}}{4x-1} = \lim_{x\to\infty} \frac{\sqrt{x^2\left(\frac{1}{x}+3\right)}}{x\left(4-\frac{1}{x}\right)} = \lim_{x\to\infty} \frac{x\sqrt{3}}{4x}$

$= \frac{\sqrt{3}}{4}$

⑬ $\lim_{x\to\infty} \sqrt{9x^2+x}-3x = \lim_{x\to\infty} \frac{\left(\sqrt{9x^2+x}-3x\right)\left(\sqrt{9x^2+x}+3x\right)}{\sqrt{9x^2+x}+3x}$

$= \lim_{x\to\infty} \frac{9x^2+x-9x^2}{\sqrt{x^2\left(9+\frac{1}{x}\right)}+3x} = \frac{x}{6x} = \frac{1}{6}$

⑭ $\lim_{x\to\infty} \sqrt{4x^2+3x}+x = \lim_{x\to\infty} \frac{\left(\sqrt{4x^2+3x}+x\right)\left(\sqrt{4x^2+3x}-x\right)}{\sqrt{4x+3x}-x}$

$= \lim_{x\to-\infty} \frac{4x^2+3x-x^2}{\sqrt{4x^2+3x}-x} = \lim_{x\to\infty} \frac{3x(x+1)}{\sqrt{x^2\left(4+\frac{3}{x}\right)}-x} = \frac{3x(x+1)}{3x}$

$= \lim_{x\to-\infty} \frac{3x^2\left(1+\frac{1}{x}\right)}{x\left(\sqrt{4+\frac{3}{x}}-1\right)} = \lim_{x\to-\infty} \frac{x}{2-1} = -\infty$

⑯ $\lim_{x\to0} \frac{1-\cos 4x}{2x} = \lim_{x\to0} \frac{2\sin^2 2x}{2x} = \lim_{x\to0} 2\cdot\frac{\sin 2x}{2x}\cdot\frac{\sin 2x}{2x}\times 2x$

H.W → 15, 17, 19 (1.3) , 3, 4 (1.3.1)

$$= 4 \lim_{2x \to 0} \frac{\sin 2x}{2x} \quad \lim_{2x} \frac{\sin 2x}{2x} \quad \lim_{x \to 0} x$$

$$= 4 \; 1.1 \; 0 = 0$$

→ (18) $\lim_{x \to 0} \frac{\sin 3x}{\tan 2x} = \lim_{x \to 0} \frac{\frac{\sin 3x}{\sin 2x}}{\cos 2x}$

$$= \lim_{x \to 0} \frac{\sin 3x . \cos 2x}{\sin 2x} = \lim_{x \to 0} \frac{\frac{\sin 3x}{3x} \; 3x \; \frac{\cos 2x}{}}{\frac{\sin 2x}{2x} . 2x}$$

$$= \lim_{x \to 0} \frac{3x}{2x} = \frac{3}{2}$$

→ (20) $\lim_{x \to 0} \frac{\cos x}{\sin x - 3} = -\frac{1}{3}$

Two Sided Limits:

$\lim_{x \to a} f(x) = L$, if and only if $\lim_{x \to a^+} f(x) = L = \lim_{x \to a^-} f(x)$

Example:- $f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8-2x & \text{if } x \leq 4 \end{cases}$, determine whether $\lim_{x \to 4} f(x)$ exists.

Ans: L.H.L :- $\lim_{x \to 4^+} \sqrt{x-4} = 0$

R.H.L :- $\lim_{x \to 4^-} 8 - 2x = 0$ R.H.L = L.H.L

$\therefore \lim_{x \to 4} f(x) = 0$, it exists.

→ ① Let $fx = \begin{cases} x^2+1 & \text{if } x < 1 \\ (x-2)^2 & \text{if } x \geq 1 \end{cases}$, @ Find $\lim_{x \to 1^+} f(x)$ an (b) does it exist?

@ Ans: L.H.L = $\lim_{x \to 1^+} x^2+1 = 1+1 = 2$

R.H.L = $\lim_{x \to 1^-} (x-2)^2 = (1-2)^2 = 1$ ∵ R.H.L ≠ L.H.L

∴ b it doesn't exist.

Next Class (Tuesday) → Quiz : Chap 1

→ ② Let $B(t) = \begin{cases} 4 - \frac{1}{2}t, & t < 2 \\ \sqrt{t+c}, & t \geq 2 \end{cases}$, Find the value of c, so that $\lim_{t \to 2} B(t)$ exists.

Ans: L.H.L = $\lim_{t \to 2^-} B(t)$ $4 - \frac{1}{2}t = 3$

$$R.H.L = \lim_{t \to 2^+} \sqrt{t+c} = \sqrt{2+c}$$

As per question, $\sqrt{2+c} = 3$ $\quad 2+c = 9 \therefore c = 7$

Continuity:

A function f is continuos at a number 'a' if (a) $f(a)$ is defined, (b) $\lim_{x \to a} f(x)$ exists, i.e. $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$, (c) $\lim_{x \to a} f(x) = f(a)$

① Test continuity:

$$f(x) = \begin{cases} x^2+1, & x < 1 \\ x+1 & x > 1 \end{cases} \quad \text{at } x = 1$$

$$R.H.L = \lim_{x \to 1^-} f(x) = x^2+1 = 2$$

$$L.H.L = \lim_{x \to 1^+} = x+1 = 2 \qquad R.H.L = L.H.L$$

$\lim_{x \to 1} f(x) = f(1) = 2$ · This function is continuos.

② $f(x) = \begin{cases} 2x+1, & x < 1 \\ 4-2x, & x \geq 1 \end{cases}$ at $x = 1$

$$R.H.L = \lim_{x \to 1^-} 2x+1 = 3$$

$$L.H.L = \lim_{x \to 1^+} 4-2x = 2$$

$R.H.L \neq L.H.L$

This function is continuous.

③ $f(x) = \begin{cases} x^2, & x < -1 \\ x, & -1 \le x \le 1 \\ \frac{1}{x}, & x > 1 \end{cases}$ at $x = 1$

Ans: (x^2, $x < -1$ doesn't match with $x = 1$, so it's out). $f(x)$ is defined at $x = 1$

R.H.L $= \lim_{x \to 1^-} x = 1$, L.H.L $= \lim_{x \to 1^+} \frac{1}{x} = 1$

$\because$ R.H.L = L.H.L $\therefore$ This function exist.

$\therefore$ This function is continuous.

$\lim_{x \to 1} f(x) = f(a) = f(1) = 1.$